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Data sheet 6AV2124-1DC01-0AX0



SIMATIC HMI KP400 Comfort, Comfort Panel, key operation, 4" widescreen TFT display, 16 million colors, PROFINET interface, MPI/PROFIBUS DP interface, 4 MB configuration memory, Windows CE 6.0, configurable from WinCC Comfort V11

General information	
Product type designation	KP400 Comfort
Display	
Design of display	TFT
Screen diagonal	4.3 in
Display width	95 mm
Display height	53.8 mm
Number of colors	16 777 216
Resolution (pixels)	
● Horizontal image resolution	480 pixel
● Vertical image resolution	272 pixel
Backlighting	
● MTBF backlighting (at 25 °C)	80 000 h
● Backlight dimmable	Yes; 0-100 %
Control elements	
Keyboard fonts	
● Function keys	
— Number of function keys	8
— Number of function keys with LEDs	8
● Keys with LED	Yes
● System keys	Yes
● Numeric keyboard	Yes
● alphanumeric keyboard	Yes; similar to mobile phone
Touch operation	
● Design as touch screen	No; without
Expansions for operator control of the process	
● DP direct LEDs (LEDs as S7 output I/O)	
— F1...Fx	8
● Direct keys (keys as S7 input I/O)	
— F1...Fx	8
● Direct keys (touch buttons as S7 input I/O)	0
Installation type/mounting	
Mounting position	vertical
Wall mounting/direct mounting	No
Mounting in portrait format possible	No
Mounting in landscape format possible	Yes
maximum permissible angle of inclination without external ventilation	35°
Supply voltage	
Type of supply voltage	DC

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Rated value (DC)	24 V
permissible range, lower limit (DC)	19.2 V
permissible range, upper limit (DC)	28.8 V
Input current	
Current consumption (rated value)	0.13 A
Starting current inrush I²t	0.5 A²·s
Power	
Active power input, typ.	3.1 W
Processor	
Processor type	ARM
Memory	
Flash	Yes
RAM	Yes
Memory available for user data	4 Mbyte
Type of output	
Info LED	Yes
Power LED	No
Error LED	No
Acoustics	
● Buzzer	Yes
● Speaker	No
Time of day	
Clock	
● Hardware clock (real-time)	Yes
● Software clock	Yes
● retentive	Yes; Back-up duration typically 6 weeks
● synchronizable	Yes
Interfaces	
Number of industrial Ethernet interfaces	1
Number of RS 485 interfaces	1; RS 422 / 485 combined
Number of RS 422 interfaces	1
Number of RS 232 interfaces	0
Number of USB interfaces	1; USB 2.0
● USB Mini B	1; 5-pole
Number of 20 mA interfaces (TTY)	0
Number of parallel interfaces	0
Number of other interfaces	0
Number of SD card slots	2
With software interfaces	No
Industrial Ethernet	
● Industrial Ethernet status LED	2
● Number of ports of the integrated switch	1; No switch integrated
Protocols	
PROFINET	Yes
Supports protocol for PROFINET IO	Yes
IRT	No
PROFIBUS	Yes
EtherNet/IP	Yes
MPI	Yes
Protocols (Ethernet)	
● TCP/IP	Yes
● DHCP	Yes
● SNMP	Yes
● DCP	Yes
● LLDP	Yes
WEB characteristics	
● HTTP	Yes
● HTTPS	No
● HTML	Yes
● XML	No

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● CSS	Yes
● Active X	No
● JavaScript	Yes
● Java VM	No
Redundancy mode	
Media redundancy	
— MRP	No
Further protocols	
● CAN	No
● MODBUS	Yes
Interrupts/diagnostics/status information	
Diagnoses	
● Diagnostic information readable	Yes; S7 controller
EMC	
Emission of radio interference acc. to EN 55 011	
● Limit class A, for use in industrial areas	Yes
● Limit class B, for use in residential areas	No
Degree and class of protection	
IP (at the front)	IP65
IP (rear)	IP20
NEMA (front)	
● Enclosure Type 4 at the front	Yes
● Enclosure Type 4x at the front	Yes
Standards, approvals, certificates	
CE mark	Yes
cULus	Yes
RCM (formerly C-TICK)	Yes
KC approval	Yes
Use in hazardous areas	
● ATEX Zone 2	Yes
● ATEX Zone 22	Yes
● IECEx Zone 2	Yes
● IECEx Zone 22	Yes
● cULus Class I Zone 1	No
● cULus Class I Zone 2, Division 2	Yes
● FM Class I Division 2	Yes
Marine approval	
● Germanischer Lloyd (GL)	Yes
● American Bureau of Shipping (ABS)	Yes
● Bureau Veritas (BV)	Yes
● Det Norske Veritas (DNV)	Yes
● Lloyds Register of Shipping (LRS)	Yes
● Nippon Kaiji Kyokai (Class NK)	Yes
● Polski Rejestr Statkow (PRS)	No
Ambient conditions	
Ambient temperature during operation	
Operation (vertical installation)	
— For vertical installation, min.	0 °C
— For vertical installation, max.	50 °C
Operation (max. tilt angle)	
— At maximum tilt angle, min.	0 °C
— At maximum tilt angle, max.	40 °C
Operation (vertical installation, portrait format)	
— For vertical installation, min.	0 °C
— For vertical installation, max.	40 °C
Operation (max. tilt angle, portrait format)	
— At maximum tilt angle, min.	0 °C
— At maximum tilt angle, max.	35 °C
Ambient temperature during storage/transportation	
● min.	-20 °C

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● max.	60 °C
Relative humidity	
● Operation, max.	90 %; no condensation
Operating systems	
proprietary	No
pre-installed operating system	
● Windows CE	Yes
configuration / header	
Message indicator	Yes
Alarm system (incl. buffer and acknowledgment)	Yes
Process value display (output)	Yes
Process value default (input) possible	Yes
Recipe management	Yes
Configuration software	
● STEP 7 Basic (TIA Portal)	No
● STEP 7 Professional (TIA Portal)	No
● WinCC flexible Compact	No
● WinCC flexible Standard	No
● WinCC flexible Advanced	No
● WinCC Basic (TIA Portal)	No
● WinCC Comfort (TIA Portal)	Yes; from V11
● WinCC Advanced (TIA Portal)	Yes; from V11
● WinCC Professional (TIA Portal)	Yes; from V11
Languages	
Online languages	
● Number of online/runtime languages	32
Project languages	
● Languages per project	32
Functionality under WinCC (TIA Portal)	
Libraries	Yes
Applications/options	
● Web browser	Yes
● Pocket Word	Yes
● Pocket Excel	Yes
● PDF Viewer	Yes
● Media Player	Yes
● SIMATIC WinCC Sm@rtServer	Yes
● SIMATIC WinCC Audit	Yes
Number of Visual Basic Scripts	Yes
Task planner	
● time-controlled	Yes
● task-controlled	Yes
Message system	
● Number of alarm classes	32
● Bit messages	Yes
— Number of bit messages	2 000
● Analog messages	Yes
— Number of analog messages	50
● S7 alarm number procedure	Yes
● System messages HMI	Yes
● System event, more (SIMATIC S7, SINUMERIK, SIMOTION, ...)	Yes
● Number of characters per message	80
● Number of process values per message	8
● Acknowledgment groups	Yes
● Message indicator	Yes
Recipe management	
● Number of recipes	100
● Data records per recipe	200
● Entries per data record	200
● Size of internal recipe memory	512 kbyte

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● Recipe memory expandable	Yes
Variables	
● Number of variables per device	1 024
● Number of variables per screen	50
● Limit values	Yes
● Multiplexing	Yes
● Structures	Yes
● Arrays	Yes
Images	
● Number of configurable images	500
● Permanent window/default	Yes
● Global image	Yes
● Pop-up images	Yes
● Slide-in images	Yes
● Image selection by PLC	Yes
● Image number in the PLC	Yes
Image objects	
● Number of objects per image	50
● Text fields	Yes
● I/O fields	Yes
● Graphic I/O fields (graphics list)	Yes
● Symbolic I/O fields (text list)	Yes
● Date/time fields	Yes
● Switches	Yes
● Buttons	Yes
● Graphic display	Yes
● Icons	Yes
● Geometric objects	Yes
Complex image objects	
● Number of complex objects per screen	5
● Alarm view	Yes
● Trend view	Yes
● User view	Yes
● Status/control	Yes
● Sm@rtClient view	Yes
● Recipe view	Yes
● f(x) trend view	Yes
● System diagnostics view	Yes
● Media Player	Yes
● HTML browser	No
● PDF display	Yes
● IP camera display	Yes
● Bar graphs	Yes
● Sliders	Yes
● Pointer instruments	Yes
● Analog/digital clock	Yes
Lists	
● Number of text lists per project	300
● Number of entries per text list	30
● Number of graphics lists per project	100
● Number of entries per graphics list	30
Archiving	
● Number of archives per device	10
● Number of entries per archive	10 000
● Message archive	Yes
● Process value archive	Yes
● Archiving methods	
— Sequential archive	Yes
— Short-term archive	Yes
● Memory location	
— Memory card	Yes

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— USB memory	Yes
— Ethernet	Yes
● Data storage format	
— CSV	Yes
— TXT	Yes
— RDB	Yes
Security	
● Number of user groups	50
● Number of user rights	32
● Number of users	50
● Password export/import	Yes
● SIMATIC Logon	Yes
Logging through printer	
● Alarms	Yes
● Report (shift log)	Yes
● Hardcopy	Yes
● Electronic print to file	Yes; PDF, HTML
Character sets	
● Keyboard fonts	
— US English	Yes
Transfer (upload/download)	
● MPI/PROFIBUS DP	Yes
● USB	Yes
● Ethernet	Yes
● using external storage medium	Yes
Process coupling	
● S7-1200	Yes
● S7-1500	Yes
● S7-200	Yes
● S7-300/400	Yes
● LOGO!	Yes
● WinAC	Yes
● SINUMERIK	Yes; with SINUMERIK option package
● SIMOTION	Yes
● Allen Bradley (EtherNet/IP)	Yes
● Allen Bradley (DF1)	Yes
● Mitsubishi (MC TCP/IP)	Yes
● Mitsubishi (FX)	Yes
● OMRON (FINS TCP)	No
● OMRON (LINK/Multilink)	Yes
● Modicon (Modbus TCP/IP)	Yes
● Modicon (Modbus)	Yes
● OPC UA Client	Yes
● OPC UA Server	Yes
Service tools/configuration aids	
● Backup/Restore manually	Yes
● Backup/Restore automatically	Yes
● Simulation	Yes
● Device switchover	Yes
Peripherals/Options	
Printer	Yes
SIMATIC HMI MM memory card: Multi Media Card	Yes; Up to 128 MB
SIMATIC HMI SD memory card: Secure Digital memory card	Yes; Up to 2 GB
SIMATIC HMI CF memory card Compact Flash Card	No
USB memory	Yes
SIMATIC IPC USB Flashdrive (USB stick)	Yes; Up to 16 GB
SIMATIC HMI USB stick	Yes; Up to 8 GB
Network camera	Yes
Mechanics/material	
Enclosure material (front)	
● Plastic	Yes

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● Aluminum	No
● Stainless steel	No
Dimensions	
Width of the housing front	152 mm
Height of housing front	188 mm
Mounting cutout, width	135 mm
Mounting cutout, height	171 mm
Overall depth	49 mm
Weights	
Weight (without packaging)	0.8 kg
Weight (with packaging)	0.9 kg

last modified: 5/27/2022

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